

Digital Manufacturing — Syllabus

Digital Manufacturing — Studio/Lab (12 Sessions) — Student-facing

Sessions: 12 × 2 hours (24 hours total)

You will design, translate to toolpaths, and fabricate with instructor support.

Sessions overview:

1. Bits Become Matter
2. CAD Foundations
3. Parametric Design
4. Material Thinking
5. Toolpaths & G-code
6. Subtractive Fab
7. Additive Fab
8. Assemblies
9. Automation & AI
10. Ecologies
11. Final Build
12. Exhibition

Assessment: Design Logbook (30%), Mini-Projects (40%), Final Project (20%), Reflection (10%).

Safety rhythm: Zero before you go. Retract before rapids. Eye protection.